

ECG Signal De-noising using Discrete Wavelet Transform

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Abstract – In this paper, wavelet de-noising method has been examined to eliminate noise from the ECG signal. Different thresholding algorithms are analyzed both theoretically and empirically. Ideal ECG signal and noise corrupted ECG signal are evaluated using MATLAB. Removal of noise because of muscle activity is difficult to handle because of the substantial spectral overlap between the ECG and muscle noise. Averaging techniques have been successfully applied to ECG signal for reduction of baseline wander noise. DWT has good ability to decompose the signal and wavelet thresholding is good in removing noise from decomposed signal. We applied wavelet transform on the input vector, thresholded it, inverse transformed it to finally achieve a signal with very low EMG noise. The analyses of thresholding techniques have been compared based on signal to noise ratio. It is observed that “rigsure” method gives optimum performance.

Keywords – Wavelet Transform, Discrete Wavelet Transform, Thresholding, ECG Signal De-Noising, SNR.

I. INTRODUCTION

ECG signals are produced from human heart activities. Potential difference between two points on the body surface, versus time is represented graphically with the help of ECG. While recording ECG in a clinical environment it is usually contaminated by baseline wandering due to respiration, power line interference and electromyography (EMG) noise. So removal of these noises is necessary in ECG analysis for correct diagnosis. Various solutions have been proposed for reduction of noise and are currently being employed in a number of systems. Wavelet thresholding de-noising method based on discrete wavelet transform (DWT) proposed by Donoho and Johnstone is often used in de-noising of ECG signal [1, 2]. Sayadi O and Brittain J.S. have used Wiener filtering and Kalman filtering methods to remove the additive noises [3, 4]. Harishchandra T. Patil gave a new method of threshold estimation for ECG signal de-noising using wavelet decomposition, where, threshold is computed using maximum and minimum wavelet coefficients at each level [5].

II. WAVELETS

Wavelets are mathematical functions with oscillatory nature similar to sinusoidal waves with the difference

being that they are of “finite oscillatory nature”. Essentially a finite length, decaying waveform, when scaled and translated results in what is called a “daughter wavelet” of the original “mother wavelet”. Hence different scaling and translation variables result in a different daughter wavelet from a single mother wavelet [6].

A wavelet transform decomposes a signal into basis functions which are known as wavelets. Wavelet transform is calculated separately for different segments of the time-domain signal at different frequencies resulting in Multi-resolution analysis or MRA. It is designed in such a way that the product of time resolution and frequency resolution is constant. Wavelet transforms [7] are classified as Continuous wavelet transforms (CWT) and Discrete wavelet transforms (DWT). The finite oscillatory nature of the wavelets makes them extremely useful in real life situations in which signals are not stationary. While Fourier transform of a signal only offers frequency resolution, wavelet transforms offer “variable time frequency” resolution which is the hallmark of wavelet transforms.

III. WAVELET BASED DE-NOISING

Wavelet de-noising methods deals with wavelet coefficients using a suitable chosen threshold value in advance. The wavelet coefficients at different scales could be obtained by taking DWT of the noisy signal. Normally, those wavelet coefficients with smaller magnitudes than the preset threshold are caused by the noise and are replaced by zero, and the others with larger magnitudes than the preset threshold are caused by original signal mainly and kept (hard-thresholding case) or shrunk (the soft-thresholding case). Then the de-noised signal could be reconstructed from the resulting wavelet coefficients. These methods are simple and easy to be used in de-noising of ECG signal.

A. Discrete Wavelet Transform

The DWT of a signal “x” is calculated by passing it through a series of filters i.e. low pass and high pass filters [8, 9]. The inner product of the signal $x(t)$ and the wavelet function $\psi_{m,k}$ provides a set of coefficients $XDWT(m,k)$ for m and k by applying DWT on signal $x(t)$. DWT can be considered as one of the multi-rate signal processing systems that use multiple sampling rates in the processing of discrete time signals. The DWT of a signal $x(t)$ is given by:

$$X_{DWT_k} = \int_{-\infty}^{\infty} x(t) 2^{m/2} \psi(2^m t - k) dt \quad (1)$$

where, $\psi_{m,k}$ is the wavelet function.

The discrete wavelet transform of a signal is calculated by passing it through a series of filters namely low pass filter and high pass filter. The coefficients associated with low pass filter is called approximation coefficients and high pass filtered coefficients are called detailed coefficients. This decomposition process is carried out until the required frequency response is achieved from the given input signal.

B. Soft and Hard Thresholding

A kind of signal estimation technique called wavelet thresholding have signal de-noising capabilities. Wavelet shrinkage operation is categorized in to two thresholding methods hard and soft. Performance of thresholding purely depends on the type of thresholding method and the thresholding rule used for the given application. In hard thresholding, the coefficients that are smaller than the threshold are vanished and the others are kept unchanged. However, the soft thresholding makes a continuous distribution of the remaining coefficients centered on zero by scaling them.

Soft thresholding [10] is given as follows:

$$\hat{x} = \begin{cases} y - \text{sgn}(y)T & \text{if } |y| \geq T \\ 0 & \text{if } |y| < T \end{cases} \quad (2)$$

Hard thresholding is given as following:

$$\hat{x} = \begin{cases} y & \text{if } |y| \geq T \\ 0 & \text{if } |y| < T \end{cases} \quad (3)$$

C. Threshold Calculating Rules

Donoho has initially proposed de-noising of signals based on fixed thresholding [10]. Here, the value of threshold (t) is computed as:

$$t = \sigma \sqrt{2 \log(n) / n} \quad (4)$$

where, $\sigma = \text{MAD} / 0.6745$

MAD represents the median of wavelet coefficients and n is the total number of wavelet coefficients.

There are four types of thresholding rules mostly used by different researchers on de-noising applications [11].

1) *Global Thresholding*: This can be considered a type of fixed threshold or global thresholding method and it is computed as:

$$T = \sigma \sqrt{2 \log(n)} \quad (5)$$

2) *Minimax Thresholding*: Minimax threshold yields minimax performance for Mean Square Error (MSE) against ideal procedures. Minimax threshold also behaves as fixed threshold. This method does the job of obtaining a minimum error between original signal and wavelet coefficients of noise signal and depending on it selects a threshold value.

3) *Rigrsure Thresholding*: It depends on the Stein's unbiased estimate of risk. In this rule, risk estimation for a particular threshold value is done. It is an adaptive

thresholding method which is proposed by Donoho and Jonstone and it is based on Stein's unbiased likelihood estimation principle [1].

4) *Heursure Thresholding*: When sure and global thresholding methods are combined together, a new rule is formed named as Heursure threshold rule. Sure estimation method becomes worthless if the signal-to noise ratio of the signal is very poor, then it will show more noises. In this kind of situation, the fixed form threshold is selected by means of global thresholding method.

D. Inverse Discrete Wavelet Transform

The IDWT of a signal "x" is calculated by passing the thresholded DWT coefficients through a series of filters i.e. low pass and high pass filters and thus, reconstructing the signal "x" from thresholded DWT coefficients. The IDWT is given by:

$$X_{IDWT}(t) = \int_{-\infty}^{\infty} \sum_{m=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} X_k^m 2^{m/2} \psi(t - k) dt \quad (6)$$

IV. PERFORMANCE EVALUATION

1) Percent Root Mean Square Difference (PRD):

One of the most difficult problems in ECG compression applications and reconstruction is defining the error criterion. The purpose of the compression system is to remove redundancy and irrelevant information. Consequently the error criterion has to be defined so that it will measure the ability of the reconstructed signal to preserve the relevant information. Since ECG signals generally are compressed with lossy compression algorithms, a way of quantifying the difference between the original and the reconstructed signal, often called distortion. The most prominently used distortion measure is the Percent Root mean square Difference (PRD) [12] that is given as follows:

$$PRD = \sqrt{\frac{\sum_{n=1}^N [x(n) - \hat{x}(n)]^2}{\sum_{n=1}^N [x(n)]^2}} \times 100 \quad (7)$$

where, $x(n)$ and $\hat{x}(n)$ are the original and reconstructed signals of length N, respectively. The PRD indicates reconstruction fidelity by point wise comparison with the original data.

2) Root Mean Square (RMS):

The root mean square [13] is calculated as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (S_{\text{original}} - S_{\text{denoised}})^2} \quad (8)$$

3) Signal to Noise Ratio (SNR):

Basically signal to noise ratio (SNR) is an engineering term for the power ratio between a signal and noise [12, 13]. It is expressed in terms of the logarithmic decibel scale.

$$SNR = 10 \log_{10} \left(\frac{E_{\text{signal}}}{E_{\text{noise}}} \right)^2 \quad (9)$$

$$SNR = 20 \log_{10} \left(\frac{E_{\text{signal}}}{E_{\text{noise}}} \right)^2 \quad (10)$$

where E_{signal} : Root mean square amplitude of the signal
 E_{noise} : Root mean square amplitude of the noise

V. EXPERIMENTS AND RESULTS

Wavelet De-noising using universal soft thresholding was performed on the ECG signals (100,102,103) taken from MIT-BIH database. The MIT-BIH Arrhythmia Database contains 48 half-hour excerpts of two-channel ambulatory ECG recordings. These are obtained from 47 subjects collected by from a mixed population of inpatients (about 60%) and outpatients (about 40%) studied by the BIH Arrhythmia Laboratory. The subjects were taken from, 25 men aged 32 to 89 years and 22 women aged 23 to 89 years. About half (25 of 48 complete records and reference annotation files for all 48 records) of this database has been freely available in PhysioNet's inception in September 1999. The 23 remaining signal files, which had been available only on the MIT-BIH Arrhythmia Database CD-ROM, were posted in February 2005. The recordings were digitized at 360 samples per second per channel with 11-bit resolution over a 10 mV range. The original ECG signal is shown in Fig. 1, 2, 3. To this input ECG signal muscle artifact is added. The Fig. 4, 5, 6 shows the signal corrupted with muscle artifact. To remove the baseline wander moving average filter was used (Fig. 7, 8, 9).

The ECG signal was decomposed at level 4 using Daubechies wavelet. Hard and soft thresholding was performed on the wavelet coefficients. Results of soft thresholding are shown in Fig. 10, 11, 12. Results of hard thresholding are shown in Fig. 13, 14, 15. It can be observed from the table I, II and III that the optimum wavelet is Db5. The performance was evaluated using SNR. We also find that though hard thresholding gives an optimum performance, we get best results using "Rigursure" soft thresholding technique. We have evaluated the results both by calculating SNR and visual inspection.

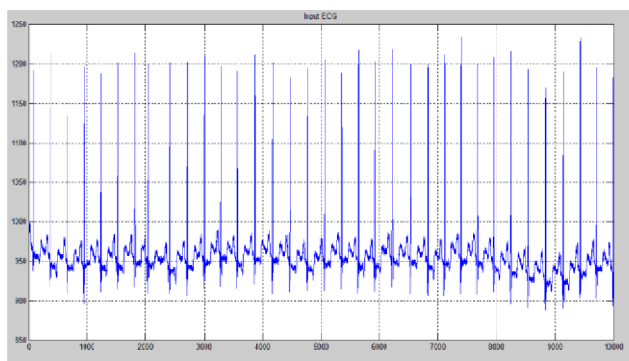


Fig.1. ECG Signal from MITBIH arrhythmia database (100)

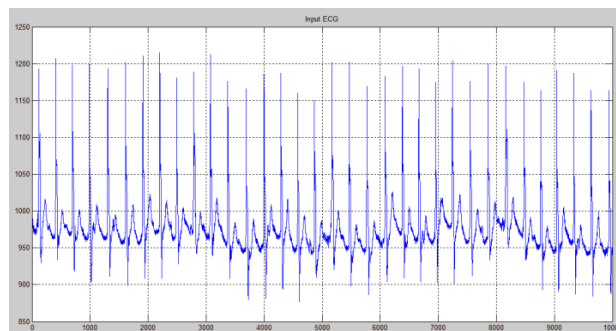


Fig.2. ECG Signal from MITBIH arrhythmia database (102)

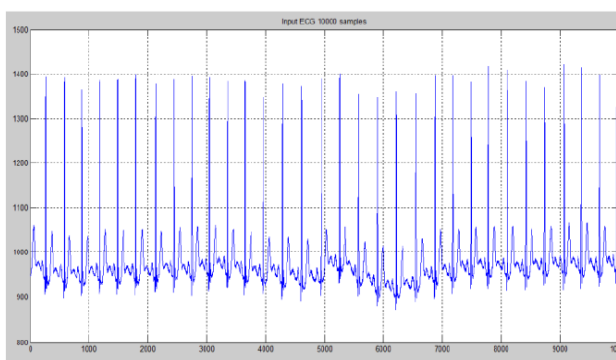


Fig.3. ECG Signal from MITBIH arrhythmia database (103)

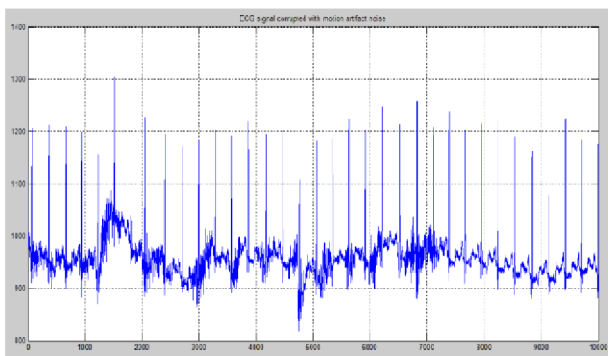


Fig.4. ECG signal (100) corrupted with motion artifact noise

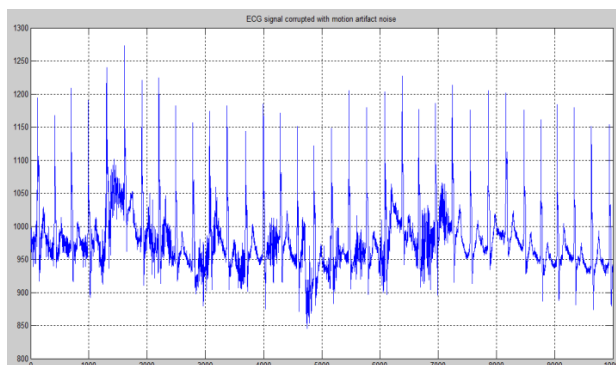


Fig.5. ECG signal (102) corrupted with motion artifact noise

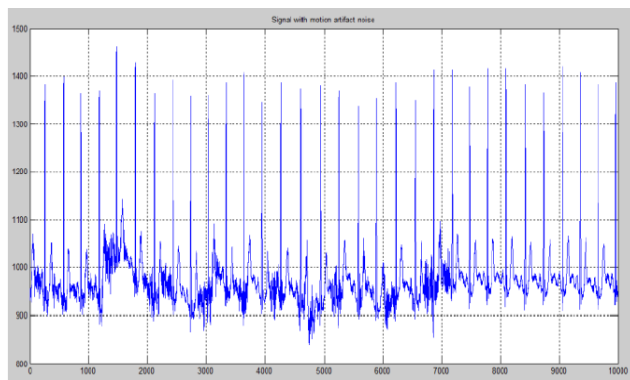


Fig.6. ECG signal (103) corrupted with motion artifact noise

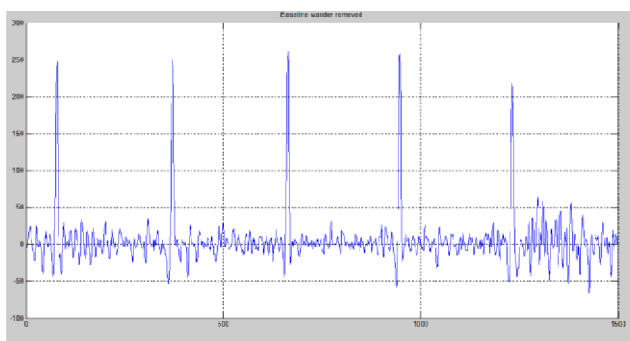


Fig.7. ECG signal (100) after removing the baseline wander using moving average filter

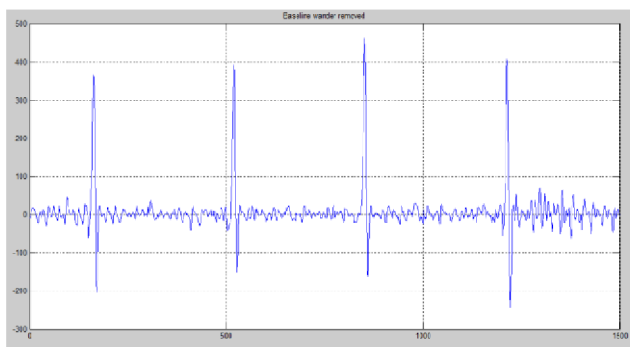


Fig.8. ECG signal (102) after removing the baseline wander using moving average filter

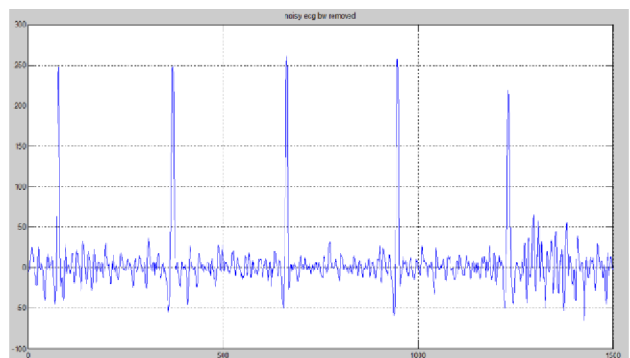


Fig.9. ECG signal (103) after removing the baseline wander using moving average filter

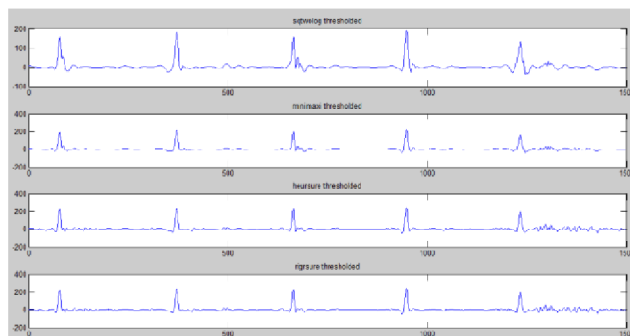


Fig.10. Soft thresholding applied to ECG 100 (MIT-BIH) using Db5 at level 4

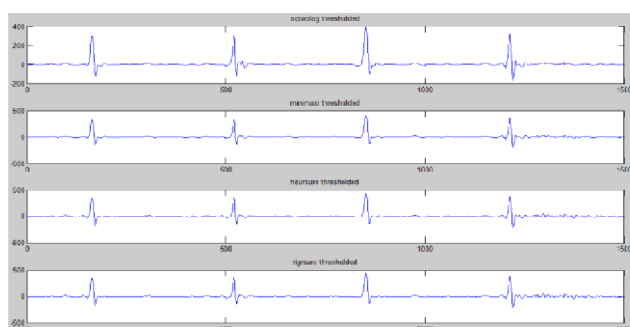


Fig.11. Soft thresholding applied to ECG 102 (MIT-BIH) using Db5 at level 4

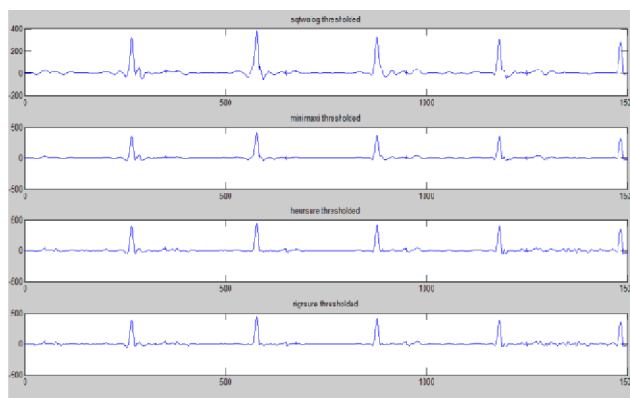


Fig.12. Soft thresholding applied to ECG 103 (MIT-BIH) using Db5 at level 4

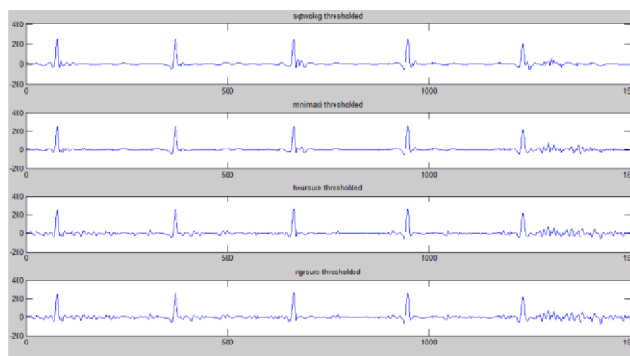


Fig.13. Hard thresholding applied to ECG 100 (MIT-BIH) using Db5 at level 4

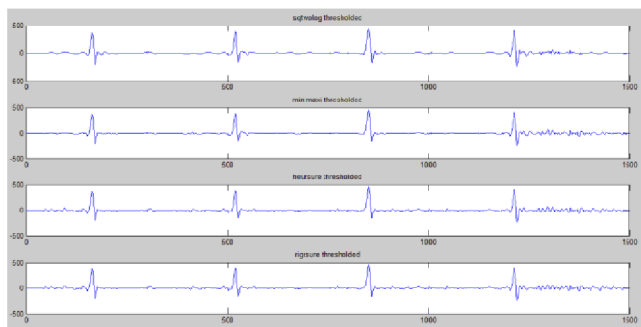


Fig.14. Hard thresholding applied to ECG 102 (MIT-BIH) using Db5 at level 4

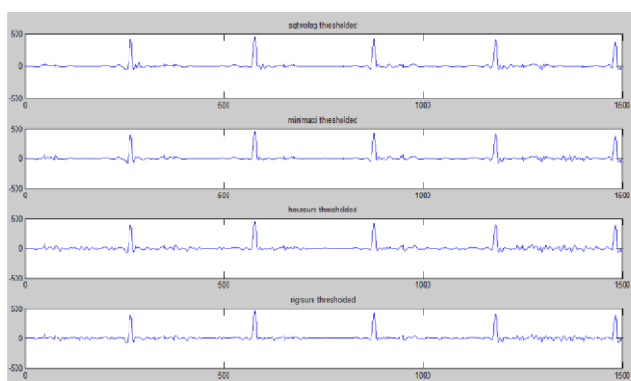


Fig.15. Hard thresholding applied to ECG 103 (MIT-BIH) using Db5 at level 4

Table I: Values of SNR using Wavelet Thresholding

Wavelet Name	Thres hold	Universal	Minimax	Huersture	Rigrsure
Db2	Hard	11.2839	12.9370	11.4949	11.4949
	Soft	8.2257	10.7906	13.3059	13.3059
Db3	Hard	12.6493	12.6807	11.4077	11.4077
	Soft	7.4646	10.4160	13.0893	13.0893
Db4	Hard	11.0762	12.1167	11.4483	11.4822
	Soft	6.9376	9.6694	12.2502	12.3017
Db5	Hard	12.2737	13.3578	11.4803	11.5072
	Soft	6.9225	10.1329	13.3828	13.4250
Db6	Hard	10.2863	12.3099	12.4139	12.4411
	Soft	6.8164	9.5132	11.4746	11.4934
Db7	Hard	9.3734	12.9534	11.4245	11.4877
	Soft	6.4390	9.2189	12.2871	12.7537
Db8	Hard	9.1527	12.2073	12.6448	12.7111
	Soft	5.6607	8.4869	11.5380	11.9517
Db9	Hard	8.7752	12.0700	11.2759	11.2792
	Soft	5.1985	8.2435	11.8797	11.8975
Db10	Hard	8.8256	11.5325	11.4382	11.5617
	Soft	5.6127	8.2552	11.5560	11.8835

Values of SNR (dB) of Wavelet Thresholding performed on ECG data 100(MIT-BIH)

Table II: Values of SNR using Wavelet Thresholding

Wavelet Name	Thres hold	Universal	Minimax	Huersture	Rigrsure
Db2	Hard	7.4108	7.7740	7.1097	7.9070

	Soft	4.9704	6.6565	7.0655	9.3261
Db3	Hard	7.3433	8.4478	8.3973	7.9378
	Soft	4.9587	7.0953	8.2562	9.2419
Db4	Hard	6.7231	7.8152	6.8902	8.0271
	Soft	4.4093	6.3240	6.5330	9.0497
Db5	Hard	7.2166	8.5250	8.1550	8.2556
	Soft	4.6411	6.5428	7.9405	9.3425
Db6	Hard	6.2481	7.7896	7.4266	7.6518
	Soft	4.3787	6.3875	6.7814	8.1037
Db7	Hard	6.7921	8.2555	8.0202	8.6303
	Soft	4.2333	6.3257	7.5191	8.9476
Db8	Hard	5.7174	8.7358	7.2765	8.2942
	Soft	3.9012	5.9446	6.1438	9.2027
Db9	Hard	5.2455	8.4387	6.5857	7.8349
	Soft	3.5223	5.5084	5.2846	7.9699
Db10	Hard	5.3404	7.9000	5.7381	7.8758
	Soft	3.6253	5.2703	4.3792	7.8545

Values of SNR (dB) of Wavelet Thresholding performed on ECG data 103(MIT-BIH)

Table III: Values of SNR using Wavelet Thresholding

Wavelet Name	Thres hold	Universal	Minimax	Huersture	Rigrsure
Db2	Hard	14.7288	15.5410	15.4933	15.4933
	Soft	10.4749	12.9113	15.6740	15.6740
Db3	Hard	14.4667	15.0519	15.3901	15.3901
	Soft	9.0817	11.8218	15.7104	15.7104
Db4	Hard	13.9872	15.9381	15.6069	15.6069
	Soft	9.5625	12.1253	16.5380	16.5380
Db5	Hard	13.9045	16.1861	15.2335	15.6238
	Soft	9.1199	11.9423	15.8242	16.5676
Db6	Hard	12.5661	14.5574	15.5318	15.5318
	Soft	7.9132	10.7060	14.5697	14.5697
Db7	Hard	11.7525	14.0691	15.5791	15.5791
	Soft	8.3412	10.7739	14.2621	14.2621
Db8	Hard	11.9658	15.0005	15.4375	15.7446
	Soft	7.5572	10.4357	14.7052	15.1813
Db9	Hard	12.3005	15.0129	14.8854	14.8854
	Soft	7.6156	10.4795	14.2219	14.2219
Db10	Hard	12.4013	14.1261	15.0395	15.3587
	Soft	7.9252	10.4436	15.2021	15.7120

Values of SNR (dB) of Wavelet Thresholding performed on ECG data 102(MIT-BIH)

VI. CONCLUSION

After experimentation it can be concluded that hard threshold not always gives better denoising performance; it depends on which wavelet thresholding algorithm was chosen. The best results were obtained using Db5 wavelet at level 4. Soft thresholding using “rigrsure” rule gave better denoising results as compared to the other rules.

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